

Changkai Zhang

Curriculum Vitae

Education History

- 2022 – 2025 **Doctor of Science**, *Ludwig-Maximilian-Universität*, München, Germany.
Research Project: Tensor Network and AI Study of the Hubbard Model.
- 2019 – 2021 **Master of Science**, *Ludwig-Maximilian-Universität*, München, Germany.
Specialized in Theoretical Physics. Grading – 1.5/1.0.
- 2016 – 2018 **Bachelor of Science**, *The University of Manchester*, Manchester, UK.
Specialized in Theoretical Physics. Grading – First Class Honours.
- 2014 – 2018 **Bachelor of Science**, *Beijing Normal University*, Beijing, China.
Specialized in Physics / Computer Science. Grading – 87/100

Publications

- 2026 **C. Zhang** and J. von Delft, *Finite-Temperature Study of the Hubbard Model via Enhanced Exponential Tensor Renormalization Group*, Phys. Rev. B **114**, 125114 (2026)
- 2025 **C. Zhang**, J.-W. Li, D. Nikolaidou, and J. von Delft, *Frustration-Induced Superconductivity in the t - t' Hubbard Model*, Phys. Rev. Lett. **134**, 116502 (2025)
- 2025 **C. Zhang** and J. von Delft, *Interpretable Artificial Intelligence (AI) Analysis of Strongly Correlated Electrons*, arXiv: 2510.26864 (2025)

Work Experience

- 2026 – now **Postdoctoral Researcher**, *The University of Hong Kong*, Hong Kong SAR, China.
Focused on application of tensor network and AI techniques in quantum lattice models.
- 2026 **Postdoctoral Researcher**, *Ludwig-Maximilian-Universität*, München, Germany.
Focused on development of open source tensor network infrastructure.

Teaching Experience

- 2022, 2025 **Teaching Assistant for Tensor Networks**
Assisted Prof. Jan von Delft in teaching the master's level course on tensor network methods, including weekly exercise sessions and support for student projects.
- 2019 **Lecture Series on Gauge/Gravity Duality**
Lecture series given at Sustech including 4 lectures on the gauge/gravity duality and its applications in holographic entanglement entropy and superconductors.

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Conferences & Workshops

- Mar, 2025 **APS March Meeting & Global Physics Summit**, Anaheim, CA, USA
Tensor Network Simulations of the Fermi-Hubbard Model
- Apr, 2025 **Workshop on Quantum Many-Body Computation**, Hangzhou, China
Frustration-Induced Superconductivity in the t - t' Hubbard Model
- Aug, 2023 **Tensor Network States: Algorithms and Applications**, Shanghai, China
Tensor Network (2D) Simulations of the Hubbard Model

Computer Skills

- Language C/C++, Python, Haskell, Mathematica, MATLAB, $\text{\LaTeX}$
- Utilities Agentic AI, Linux & CLI tools, HPC Operations
- Algorithm Tensor Networks, Transformers, Diffusion Models & LoRA

Languages

- Chinese First language, simplified & traditional
- English Second language, oral & written, daily & academic
- Deutsch Third language, beginner's level, oral & daily

Interests & Hobbies

- Aviation All sorts of model aircraft & drones, including fixed-wing aircraft, helicopters and rockets. Also interested in commercial flight safety.
- Network A web front-end design, back-end management & development, including several cloud computing instances running various web services.
- Music & Art Chinese traditional-style music & traditional instruments. AI-generated art & music, e.g. stable diffusion, voice synthesis and cloning etc.
- Animation 3D animation design & rendering via Blender, including character modeling, rigging, animation and scene design for videos and gaming.
- AI Agent Retrieval-augmented (local) AI agent systems for managing computerized knowledge and creating efficient human-computer interfaces.

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